ROLL NO: | | | | | | | | | | |

(To fill the roll no is compulsory)

INVERTIS UNIVERSITY BAREILLY

EVEN SEMESTER EXAMINATION - 2024-25

M. Tech. (CSE)

Semester: II

SUBJECT CODE: MCS203

MOBILE COMPUTING

Time: 03:00 Hrs. **Maximum Marks: 70**

NOTE: THE QUESTION PAPER CONTAINS THREE SECTIONS. ATTEMPT ALL THE SECTIONS.

SECTION – A

NOTE: Attempt all parts of the following questions. (14 Marks)

Q. 1. Attempt all questions (1×7 = 7 Marks)

(i) What does GSM stand for?

(ii) Name one key challenge in mobile ad-hoc networks (MANETs).

(iii) What is handover in the context of mobile communication?

(iv) What is the purpose of a SIM card in a mobile device?

(v) Name one wireless technology commonly used for short-range communication.

(vi) What is the basic function of a mobile IP?

(vii) What does SMS stand for?

Q. 2 Attempt all questions (1×7 = 7 Marks)

(i) Name one layer in the protocol stack of mobile communication systems.

(ii) What is the concept of location-based services (LBS)?

(iii) What is the role of a base station in cellular networks?

(iv) What is the function of a mobile switching center (MSC) in a cellular network?

(v) What is Wi-Fi Direct?

(vi) Name one security concern specific to mobile devices.

(vii) What is the role of push notifications in mobile applications?

SECTION – B

NOTE:Attempt all questions. (7×2×2 = 28 Marks)

Q.3.(i)(a) How does mobility affect data replication when we consider replication on mobile platform?

OR

(b) Explain clustering. What are the requirements and application areas of clustering?

(ii)(a) What is mobile agent? Discuss the server architecture with the help of diagram. Also, list the security threats to a mobile agent system.

OR

(b) Discuss the schemes of mobile transaction management. List out various issues concerned to transaction management.

Q.4.(i)(a) What are the advantages and disadvantages of using a wireless transmission as compared to a fiber or wire transmission?

OR

(b) Name the main differences between Adhoc network and other network. What advantages do Adhoc networks offers? Explain in detail by giving suitable example.

(ii) (a) Compare table driven and on- demand routing protocols.

OR

(b) Discuss the architecture of GSM. Also explain the security issues to implement GSM.

SECTION – C

NOTE: Attempt all questions. (14×2 = 28 Marks)

Q.5 (a) Draw a diagram showing the positioning of wireless network via wired networks. Why is a wired network usually part of wireless infrastructure?

OR

(b) Describe various multiplexing techniques. Explain why a given bandwidth is used most efficiently in CDMA.

Q.6. (a) "CSMA/CD is not suitable protocol for wireless LAN". Give reason in favor of or against the statement.

OR

(b) What is cellular system? Write the advantages and drawbacks of keeping small cells.